

Mercury

Rocky, crater-covered Mercury is the smallest planet and the closest to the Sun. It has the thinnest of atmospheres, and conditions on its surface are extremely harsh. Although visible to the naked eye, Mercury is never far from the Sun and so is often difficult to find.

FEATURES

Mercury is very dense compared to the other rocky planets, and beneath its rocky crust and mantle is a large iron core. It is not certain whether this is because the region of the solar system where Mercury was formed was rich in iron, or whether the early cratering of Mercury has eroded its rocky mantle. The core formed when the heavy iron sank within the young planet.

The planet's surface is pockmarked by thousands of craters—from small, bowl-shaped ones to the vast Caloris Basin (right). Meteorite bombardment churned up the surface and produced a powdery soil that reflects little light and is very dark, just like lunar soil.

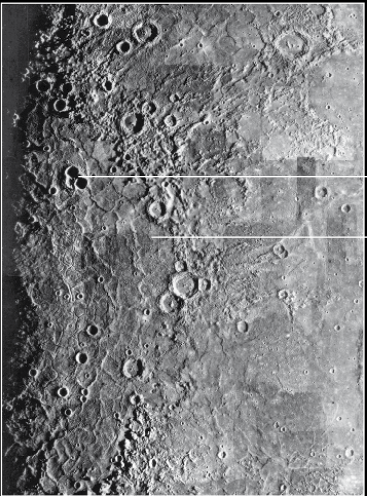
Mercury is too small and too hot to have anything more than a very thin atmosphere. This is either captured from the solar wind of gas constantly escaping the Sun or produced by the roasting of surface rocks. Much escapes in the daytime, and is constantly replenished.

Mercury has the most elongated planetary orbit: its distance from the Sun varies between 28.6 million miles (46 million km) and 43.4 million miles (69.8 million km), and it spins around exactly three times during every two orbits. This makes a “day” on Mercury last 176 Earth days.



Cratered world

Mercury is a dry, rocky world reminiscent of Earth's Moon. It has a very old surface, much of which is very heavily cratered, rather like the lunar highland. The remaining area is younger; it consists of more lightly cratered plains of solidified volcanic lava, somewhat similar to the lunar maria.

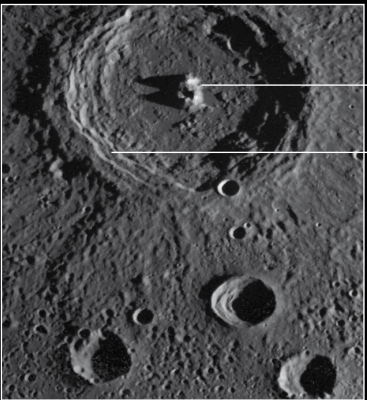


more recent impact craters

lava-filled basin floor

Caloris Basin

The huge, multiringed Caloris Basin impact crater is about 840 miles (1,350 km) across (its center lies to the left of this image). The asteroid that caused it was about 60 miles (100 km) wide. The resulting seismic waves traversed Mercury, shattering the surface on the other side, before traveling back to cause fracturing and landslides.



mountainous peak 1.8 miles (3 km) high

walls have slipped, forming terraces

Brahms Crater

This bowl-shaped crater is 60 miles (97 km) across and was formed when an asteroid hit Mercury about 3.5 billion years ago. It has a prominent central mountainous peak and inner walls that have slipped inward due to structural weakness and the influence of gravity.

OBSERVING THE TWO PLANETS

Mercury and Venus are both inside Earth's orbit, so they never appear far from the Sun. Mercury is seen closer to the Sun than Venus, which orbits at a greater distance. As a result, the two are only seen in the early morning eastern sky before the Sun rises, and in the western evening sky after the Sun has set. Mercury is only visible for around two weeks when the planet is at elongation. Venus can be very prominent at elongation and is then often by far the brightest body in the sky.

Dates of elongation are included in the Special Events listings of the Monthly Sky Guide (pp.44–69).



Naked-eye view

Here, the Sun is below the horizon and is illuminating the sky near the horizon. The brighter of the circled planets is Venus. Mercury is fainter because it is both smaller and has a rocky surface that reflects less sunlight. Few stars are seen in the vicinity because their light has been absorbed by Earth's atmosphere. Both planets go through a phase cycle similar to the Moon's, but these are only seen with optical aids.

